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An integrated study of fill and deformation in the Andean intermontane basin of Nabón (Late Miocene), southern Ecuador

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Abstract

The Nabón basin is part of several Miocene intermontane basins related to Andean uplift, continental arc formation and coeval deformation during subduction of the Nazca plate under the South American continent. The basin sequence consists of fluvial, alluvial fan, lacustrine, and pyroclastic mass-flow deposits corresponding with intereruptive and syneruptive stages of basin-fill history. Combined zircon fission-track and palaeomagnetic measurements suggest that the series lies within the 4r chron (ca. 8.5–7.9 Ma). It overlies unconformably the ignimbrite basement dated at ca. 26–19 Ma. Synsedimentary tectonic deformation in the basin is manifested by a master reverse fault along the western margin, differential basement block uplift and related sedimentary bedding geometries such as cumulative sedimentary wedges. The spatial orientation of the tectonic and sedimentary features indicates that the basin underwent WNW–ESE maximal shortening during the fill stage. The short-lived sediment-fill history may be explained, either by subduction-related contractional pulses also recognized in the Peruvian and Colombian Andes or, by the particular position of the basin in the central axial part near the weak fault of a regional translational system.

1. Andes of Ecuador

The continental volcanic arc of the Cordillera Occidental and Real of the Ecuadorian Andes (Fig. 1) was formed during the Oligocene to Present after the accretion of the Cretaceous–Palaeogene oceanic Piñon–Macuchi terrane (e.g. Feininger and Bristow, 1980), which comprises the basement of the Cordillera Occidental and the Costa areas (Fig. 1a). The driving mechanism for consequent uplift, volcanism and deformation in the Andean Cordilleras was the subduction of the Nazca plate beneath continental South America (e.g. Lebrat, 1985; Bourgois et al., 1990). Obliquity of subduction was accommodated by dextral transcurrent movements along NNE-striking faults in the South American plate. Subduction-related calc-alkaline volcanic activity was more or less continuous (Fig. 2; Barberi et al., 1988; Lavenu et al., 1992) from the Oligocene to Present. However, there are significant differ-

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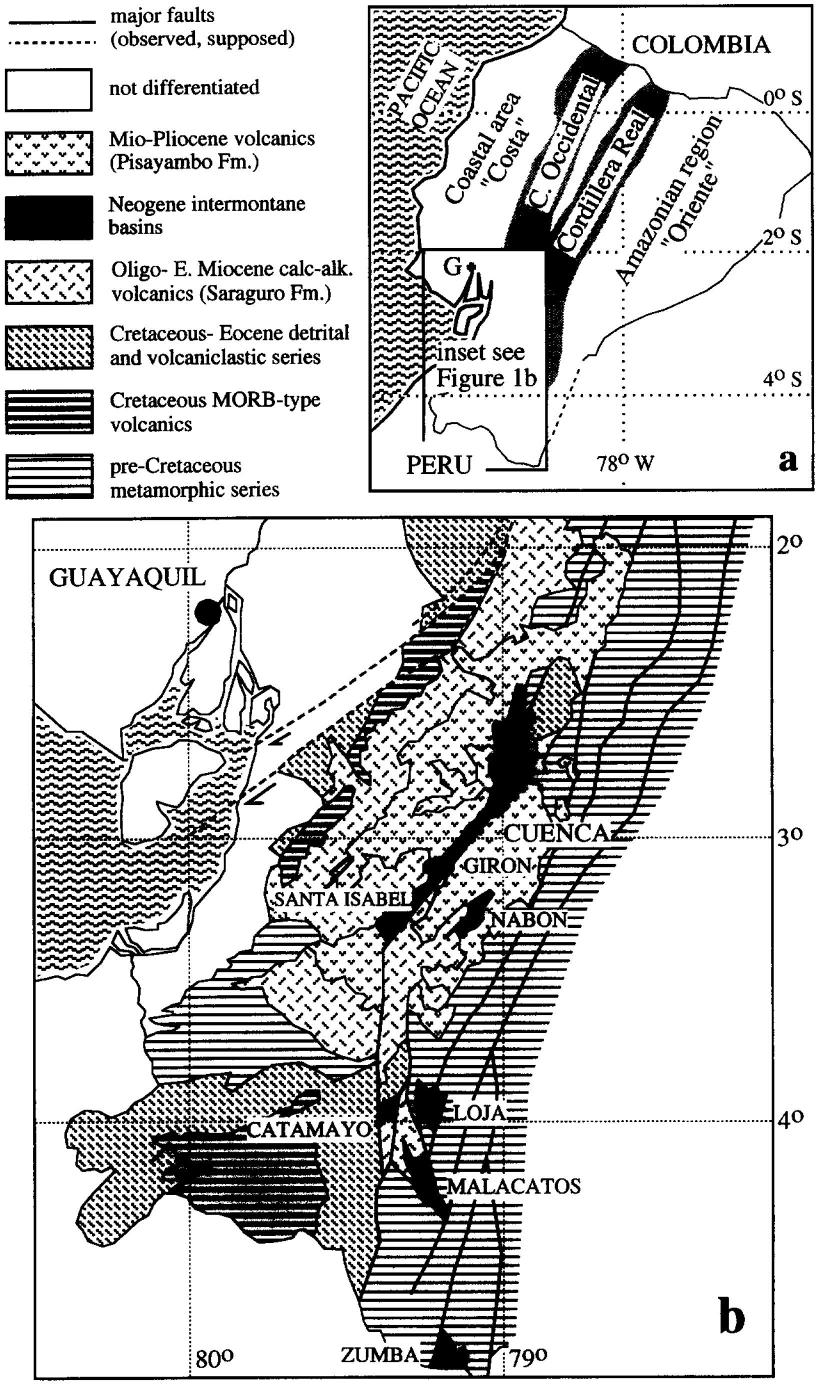


Fig. 1. Simplified maps of Ecuador. (a) Morphotectonic areas of Ecuador with Pacific Nazca plate (for more details, see e.g. Baldock, 1982). (b) Geological map of southern Ecuador with position of the main Neogene intermontane basins (prepared after Litherland et al., 1993).

ences in timing and intensity of volcanic activity between the northern and southern Ecuadorian Andes. These differences are thought to correlate with the north–south shallowing of the subduction angle of the Nazca plate beneath the South American plate (Pennington, 1981).

2. Geological setting of continental intermontane basins

Continental intermontane basin formation is a common feature of the South American Andes. This process accompanies the uplift and deformation of the continental arc, but in general, the timing and rate are poorly known. In the southern part of Ecuador where the two main Cordilleras converge, several continental intermontane basins formed during the Neogene (Fig. 1b). These are the large Cuenca–Girón–Santa Isabel basin complex, the Nabón basin to the east and several basins (Loja, Malacatos, Zumba) to the south. The basin-fill series consist both of primary and reworked material derived from coeval volcanoes and adjacent metamorphic basement rocks. Except for the Cuenca basin, the stratigraphy of the different basins is poorly constrained (e.g. Bristow, 1973; Kennerley, 1973;

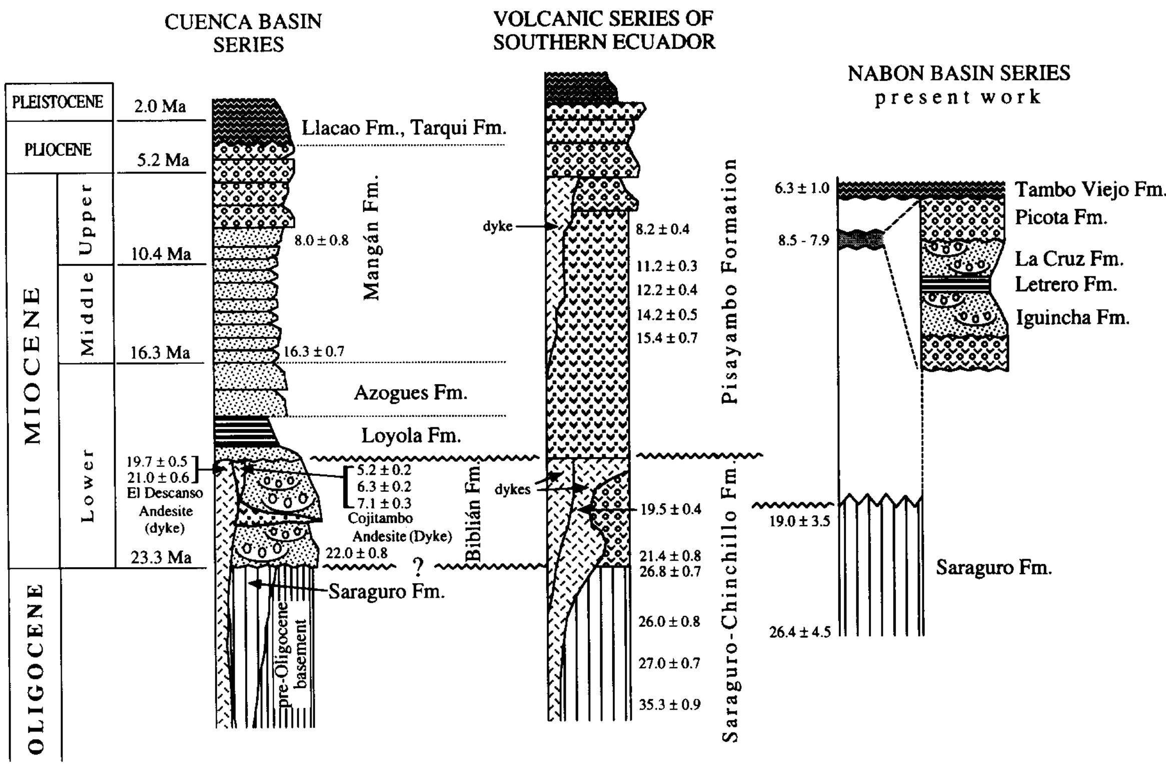
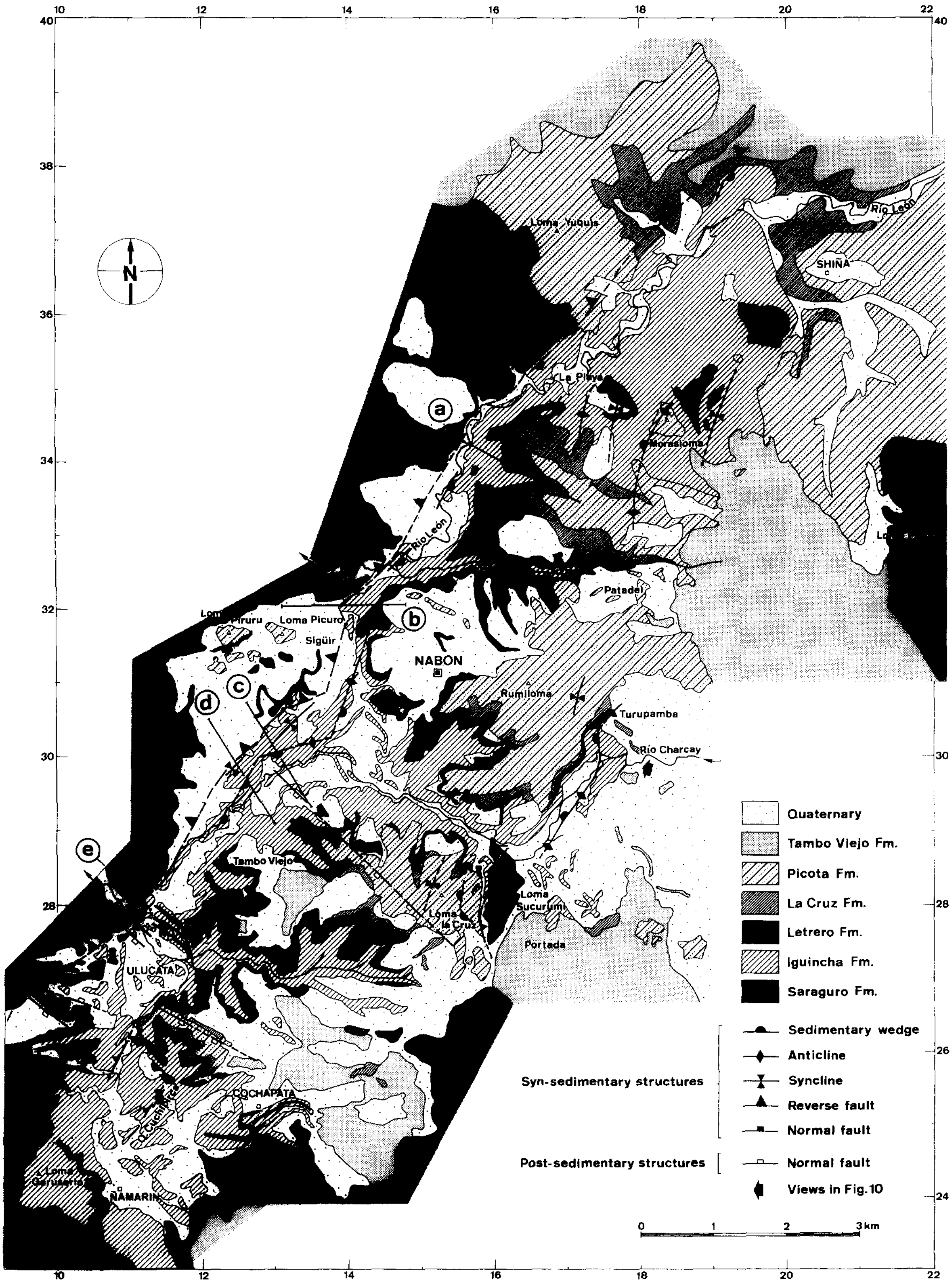


Fig. 2. Stratigraphic compilation of the Neogene volcanic continental arc series in southern Ecuador (Litherland et al., 1993), the basin-fill series of Cuenca (Noblet et al., 1988) and Nabón (present work). The continental volcanic arc sequence in southern Ecuador, preceding and paralleling the intermontane basin formation is subdivided into two main tracts. The lower, the Saraguro–Chinchillo Fm. (Kennerley, 1973; Bristow, 1973), consists of thick andesitic lava flows and breccias overlain by acidic primary and reworked volcaniclastic series. The upper part consists of rhyolitic volcaniclastics (tuffs, debris flows, air fall deposits) which were attributed earlier to the Tarqui Fm. of Quaternary age (Bristow, 1973; Baldock, 1982). However, radiometric measurements revealed Middle to Late Miocene ages and the name Pisayambo Fm. was introduced by Litherland et al. (1993); see also Fig. 1b. This upper series includes also the thick volcanogenic mud flows of probably Pliocene age. No Quaternary volcanic activity has been detected here so far. K–Ar radiometric ages after Kennerley (1980), Barberi et al. (1988), Lavenu et al. (1992) and Rivera et al. (1992). Age data from the Nabón basin-fill series are discussed later. Symbol key as in Fig. 4.


N
Loma Yuquis
Río León
SHIÑA
La Playa
Morasloma
Patadel
Loma Piruru
Loma Picuro
Sigüir
NABON
Rumiloma
Turupamba
Río Charcay
Tambo Viejo
Loma la Cruz
Loma Sucurumi
Portada
ULUCATA
COCHAPATA
Loma Garuseria
ÑAMARIN
a
b
c
d
e
Quaternary
Tambo Viejo Fm.
Picota Fm.
La Cruz Fm.
Letrero Fm.
Iguincha Fm.
Saraguro Fm.
Syn-sedimentary structures
Sedimentary wedge
Anticline
Syncline
Reverse fault
Normal fault
Post-sedimentary structures
Normal fault
Views in Fig.10
0 1 2 3km

Bristow and Parodiz, 1982; Noblet et al., 1988). Synsedimentary tectonic activity appears to have affected the basin-fill series (Lavenu and Noblet, 1989; Winkler et al., 1993). The period of intermontane basin formation probably coincides with accelerated Nazca plate subduction between 26 and 6 Ma (with a peak between 20 and 10 Ma) or since 26 Ma as concluded from kinematic plate reconstructions (Pilger, 1983; Pardo-Casas and Molnar, 1987; Daly, 1989).

The Cuenca basin with a sediment fill of approximately 4200 m thick has been investigated earlier in some detail (Noblet et al., 1988; Egüez and Noblet, 1988; Noblet and Marocco, 1989, see a compilation in Fig. 2). According to these authors, the volcanic Saraguro Fm. and pre-Oligocene metamorphic rocks are overlain by braided river and alluvial fan deposits of variable thicknesses (Biblián Fm., Lower Miocene) reflecting strong synsedimentary tectonic activity during the period from 25 to 20 Ma. In a marginal basin position coeval igneous dykes and ignimbrite flows (e.g. the "El Descanso andesite" of ca. 20 Ma, Kennerley, 1980) formed. These earlier formations are covered unconformably by lacustrine sediments of the Loyola Fm. followed by deltaic, fluvial and turbiditic lacustrine series of the Azogues Fm. The overlying Upper Miocene to Pliocene (?) Mangán Fm. consists of fluvial and alluvial fan deposits, the latter (Turi Fm. of Bristow, 1973) passing discordantly across the basin margins. Finally, the main basin-fill series is topped by the volcanic sequence of the Plio–Pleistocene (?) Llacao Fm. Noblet et al. (1988) have recognized continuous but variably oriented synsedimentary tectonic deformation in the Cuenca basin related to major N–S- and NE–SW-trending fault zones working as strike-slip or reversed faults depending on the changing orientation of the principal stress field. They suggest synsedimentary NNE–SSW-directed contraction for the Early Miocene and E–W-directed contraction for the Late Miocene to Pliocene.

The Nabón basin is an elongate NNE–SSW-striking structure covering about 120 km^2 (Fig. 1b) situated 60 km south of Cuenca. It is surrounded by Oligocene to Miocene volcanic series (Saraguro and Pisayambo Fms.). The dominantly flat-lying basin-fill series discordantly overlies the volcanic basement (Saraguro Fm.) and is 500–600 m thick. The fill sequence was mainly supplied from coeval volcanic centres and to a minor extent from metamorphic rocks of the Cordillera Real to the east (Fig. 1a). The fill is thin compared to that in the nearby Cuenca basin. The western boundary between the basin-fill series and the adjacent volcanics is morphologically sharp. The eastern margin is rather smooth, due partly to depositional overstepping by the basin infill, and partly to sealing with young post-fill volcanics. The greatest thickness of basin fill and best field evidence of tectonic deformation is found along the western margin. We are able to show that this deformation was syndepositional because of folding, faulting and differential uplift which caused discordances and fanning of the layers away from the generating tectonic structures (cumulative sedimentary wedges). The deformation pattern is used to reconstruct the orientation of the tectonic strain field. To calibrate the synsedimentary tectonic movements in time, a detailed stratigraphic frame was made using zircon fission-track and palaeomagnetic measurements. These data combined with sedimentological analysis allows the first detailed picture of the Nabón basin history to be constructed. However, the lack of precise information about other intermontane basins (with the exception of the Cuenca basin) limits the certainty of correlation with the subduction kinematics of the Nazca plate along the Pacific coast.

3. The Nabón basin-fill series

The sediment series of the Nabón Fm. was described and divided into three units (DGGM,

Fig. 3. Geological map of the Nabón basin (after Hungerbühler, 1993; Steinmann, 1993; A. Egüez, unpubl. data). Traces of geological profiles in Fig. 9 are indicated. Early and late fill phase (synsedimentary structures) and post-fill phase (postsedimentary structures) are distinguished after the relative age of the sediments involved in tectonic deformation.

1973, 1974; Bristow, 1976). Based on lithostratigraphic comparisons with other intermontane basins, Bristow (1976) assumed a Late Miocene age. The present geological mapping (Fig. 3) resulted in a detailed lithostratigraphic and sedimentological scheme for the Nabón Group (Hungerbühler, 1993; Steinmann, 1993). A north–south-oriented series of composite profiles has been constructed (Fig. 4). In the southern part, the sediment-fill series is reduced, partly because of non-deposition, partly by erosion.

The basement of the basin is the Oligocene–Early Miocene *Saraguro Fm*. It consists mainly of ignimbrites (with numerous columnar cooling structures) and a few andesitic lava flows. It outcrops extensively along the western basin margin. At the eastern margin only a few uplifted blocks are present. The volcanic Saraguro Fm. covers a large area in southern Ecuador and is at least 1000 m thick (Baldock, 1982).

The *Nabón Group* unconformably overlies the Saraguro Fm. and is composed of reworked (intereruptive) and primary (syneruptive) volcaniclastic deposits of about 550 m thickness. In the southern part, pre-fill phase horst–graben structures in the basement are filled and overstepped by the oldest basin-fill formations. We have subdivided the group from base to top as follows.

The Iguincha Formation

The Iguincha Formation, about 200 m thick, is characterised by strong lateral and vertical variations. Several distinct members can be defined. The *Infiernillo Member* (type locality grid refer-

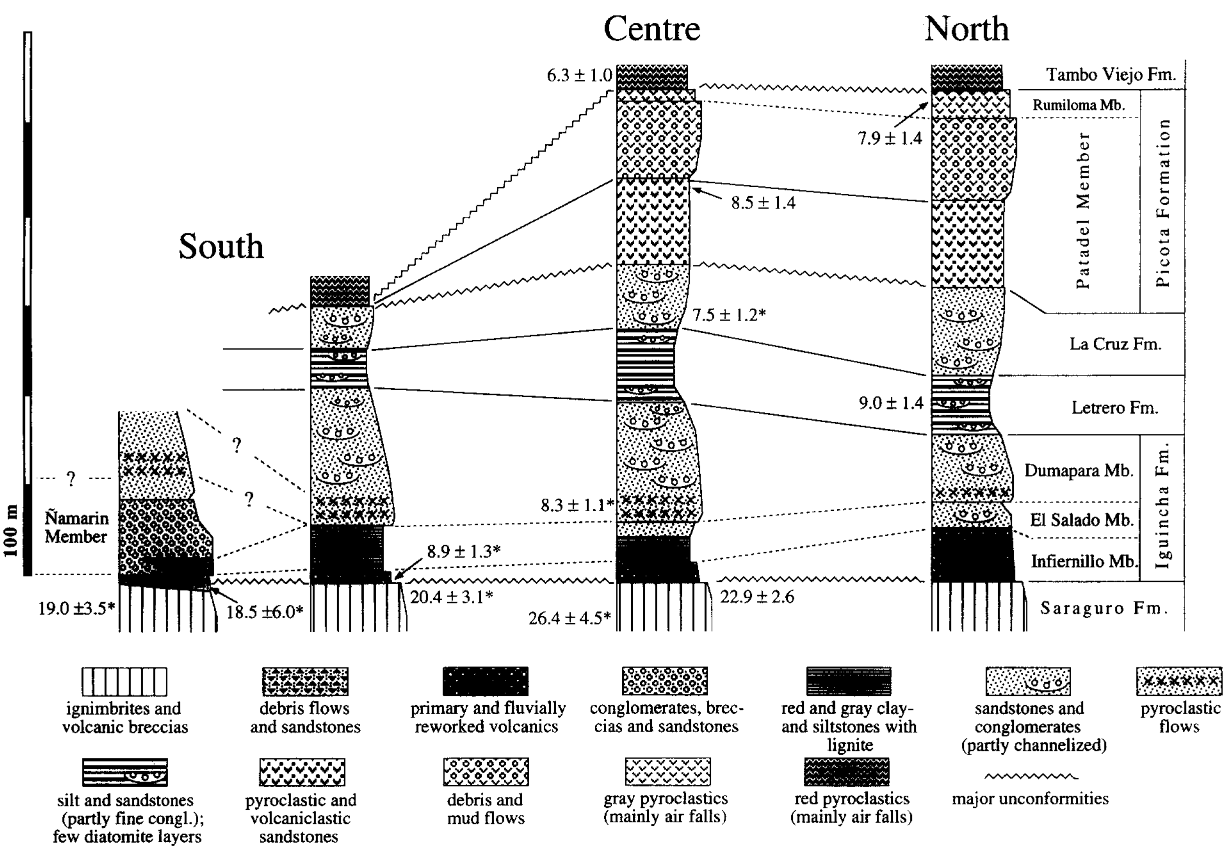


Fig. 4. Composite lithostratigraphic sections from the northern, central and southern part of the Nabón basin. Zircon fission-track ages with 2σ error are indicated. In the case of detrital contamination of the zircon samples (*) the depositional age (Table 1) is indicated.

ence: 11.115/27.840, IGM, 1970, 1971; see Fig. 3) consists of primary and aquatically reworked volcaniclastic breccias, sandstones and siltstones with a typical admixture of reworked, dark, metamorphic pebbles derived from the Cordillera Real. Volcaniclastics (ash and pumice beds) are commonly reworked in slumps, debris flows, mud flows and partly in small river channels. In the northern part of the basin, the member is about 100 m thick and thins towards the south. The high content of primary volcanic material suggests that it is the product of nearby volcanic activity (syneruptive stage of basin fill).

In the overlying *El Salado Member* (12.550/29.250), the rate of fluvial reworking and input into the basin increased considerably. In the north, coarse sandstones and channelized pebbly sandstones prevail, while to the south there is a transition to thin-bedded fluvial sandstones and laminated red and grey siltstones. The siltstones contain lignite layers and a rich leaf flora. The correlation of the lignite layers with similar formations in the Loja intermontane basin induced Bristow (1976) to suggest a Late Miocene age for the Nabón basin fill.

The southeastern margin is dominated morphologically by the 100 m thick *Ñamarin Member* (10.900/23.700), composed of debris and sheet flows with primary volcanic intercalations deposited on alluvial fans entering the basin from the east. In the Ñamarin area, anoxic lake sediments of the El Salado Mb. are intercalated between branches of an alluvial fan.

The 50–150 m thick *Dumapara Member* (13.400/27.100) consists of a vertically variable succession of channelized conglomerates, sandstones and siltstones deposited in a bedload-dominated fluvial system with overbank tracts. In the western basin it overlies conformably the El Salado Mb., in the central eastern part a lateral transition with the Ñamarin Mb. alluvial fan is observed. In the western part at the base of the Dumapara Mb. over a thickness of 20–30 m there are numerous pyroclastic flows intercalated which point to another short pulse of volcanic input.

The Letrero Formation

The Letrero Formation (13.600/28.740) is composed of lacustrine sediments with strong fluvial clastic input. The series contains siltstones and fine-grained sandstones (partly wave-rippled) with a few diatomite layers in the central part of the basin, where a maximum thickness of about 80 m can be observed. A general coarsening trend of the sediments towards the eastern, southern and northern margins is noteworthy.

The La Cruz Formation

The La Cruz Formation (13.950/28.450) is a discontinuous formation (max. 80 m thick) occurring mainly in the central and northern parts of the basin, where it concordantly overlies the Letrero Fm. The La Cruz Fm. consists of a thick-bedded, white, siltstone succession with local conglomeratic channel-lag and point-bar sediments, which were deposited in mixed-load-dominated fluvial systems. The rivers entered the basin at the central eastern and northeastern margins; in the former entry area (Portada) a meander channel (more than 100 m wide) is exposed. In the northern part of the basin the Dumapara Mb., Letrero Fm. and La Cruz Fm. are difficult to separate, because continuous strong clastic input has partly obliterated the stratigraphy farther to the south.

The Picota Formation

The Picota Formation represents at least a 200 m thick, wedge-shaped, volcaniclastic body which thins rapidly from NE to SW. It unconformably overlies the Letrero and La Cruz Fms. The lower *Patadel Member* (17.690/31.800) contains coarse volcaniclastic debris and mud flows intercalated with air fall and pyroclastic flow beds. The *Rumiloma Member* above (16.550/31.020) concordantly overlies the older coarse volcaniclastics and consists mostly of horizontally layered and laminated ash fall deposits. Similar rocks are also present outside the basin (e.g. at the Loma Picuro and Loma Piruru) where they discordantly overlie the Saraguro Fm. Over all, the Picota Fm. was deposited as a coarsening-upward volcaniclastic alluvial fan (sensu Smith, 1991) and indicates a strong eruptive phase in the basin-fill source area.

The Tambo Viejo Formation

The Tambo Viejo Formation (13.450/31.800) covers extensively the eastern basin margin and

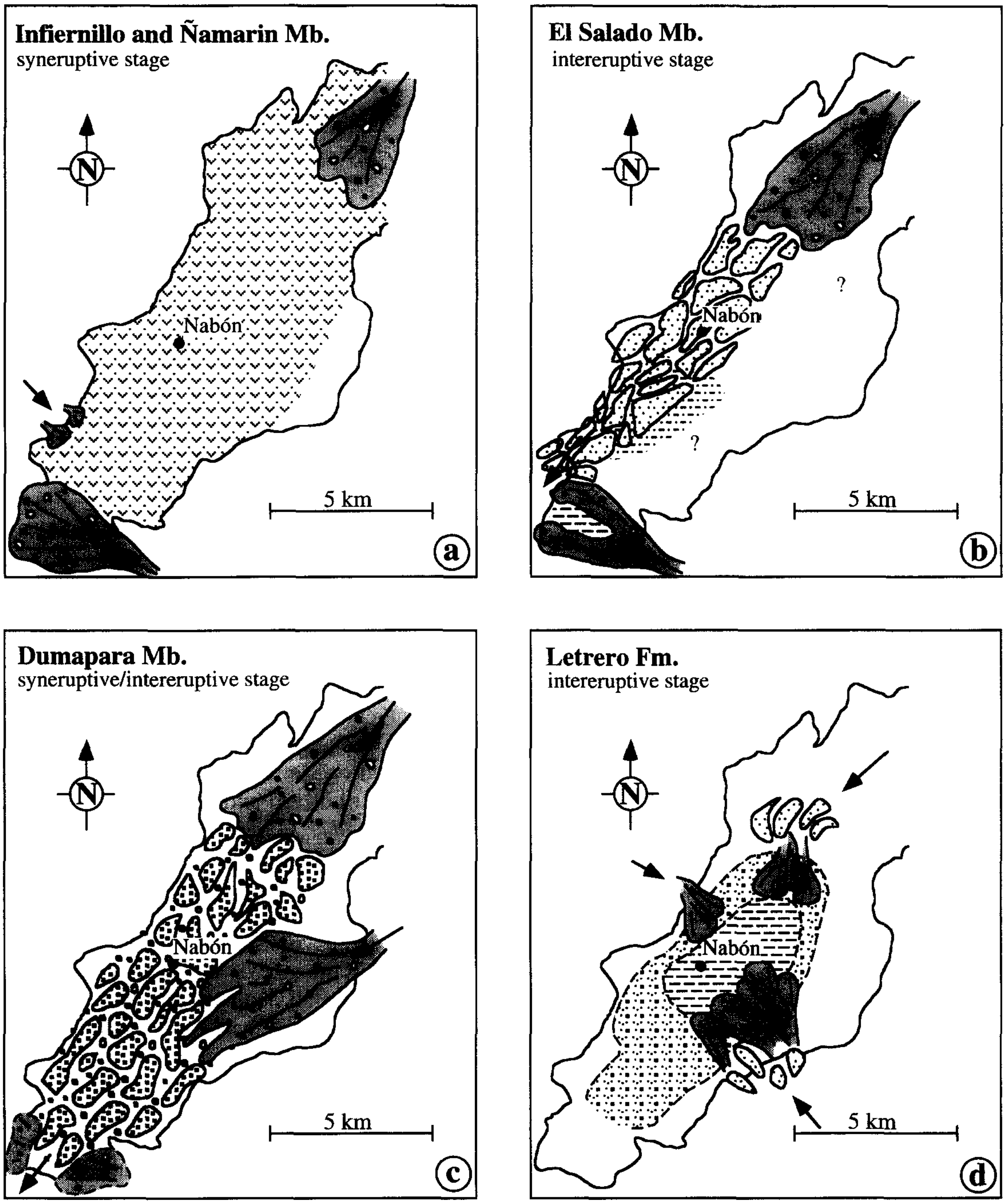


Fig. 5. Sketch of the Nabón basin sediment-infill history following lithostratigraphic subdivisions. Syneruptive and intereruptive stages refer to the intensity of volcanic activity in the area adjacent to the basin. During syneruptive stages the basin fill is dominated by primary volcanic deposits; during intereruptive stages the sediment fill occurred by aquatic processes.

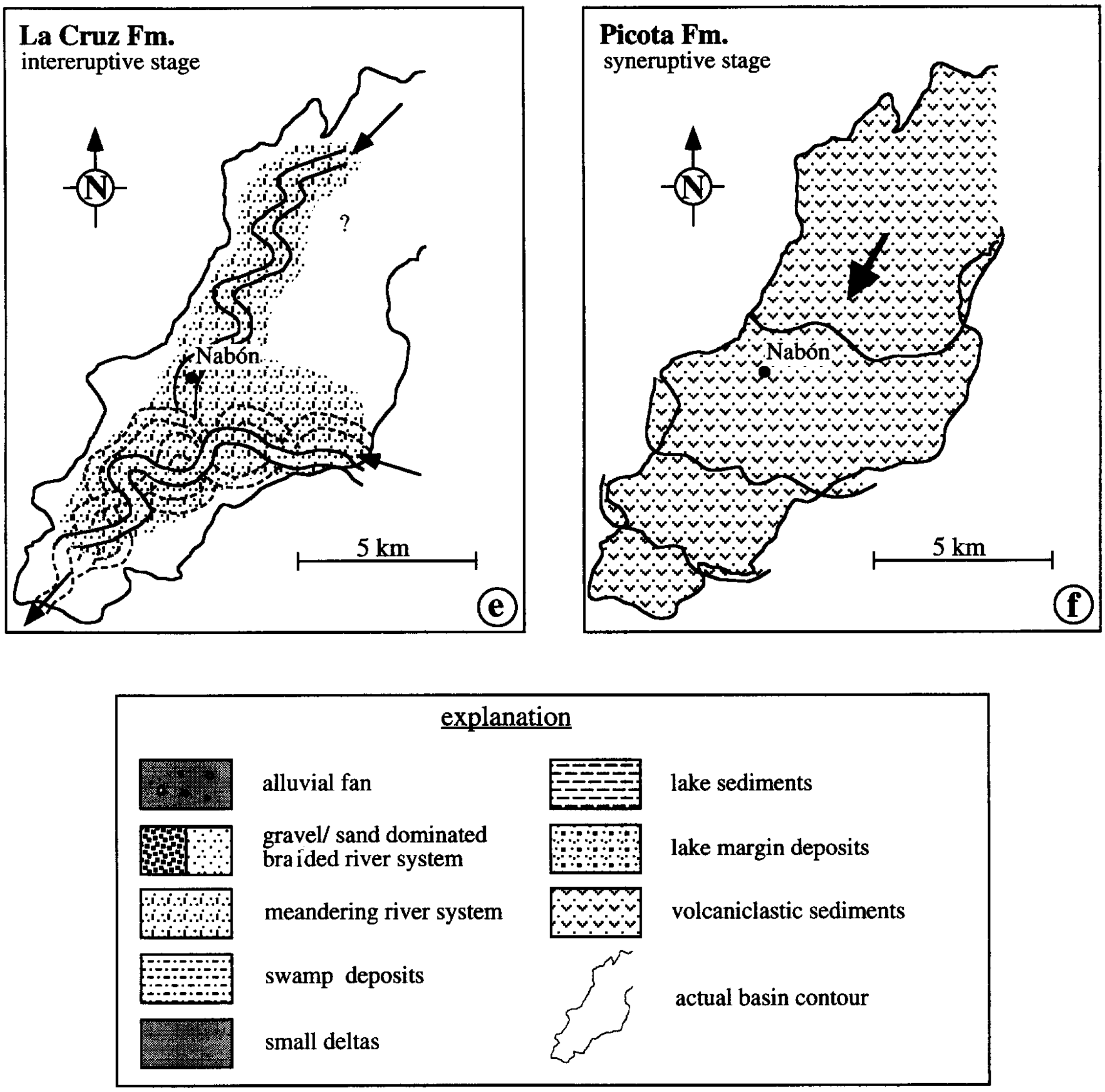


Fig. 5 (continued).

extends from there as a tongue-shaped thin (max. 10 m) sheet to the Tambo Viejo plateau. The formation is characterised by red-weathered, volcanic, air fall deposits and discordantly overlies the basin infill and basement adjacent to the basin. It represents the youngest volcanic deposits in the area and correlates with younger parts of the Pisayambo Fm. (Fig. 2 and Litherland et al., 1993).

4. Sedimentary evolution

The facies pattern of the basin-fill series shows strong lateral and vertical variations, typical of volcaniclastic sedimentary environments where the intensity of coeval volcanic activity changed in time (Smith, 1991). During sedimentation, basin drainage occurred in a general axial SSW direction and the outlet was situated at the southern termination of the basin remnant. Great parts of the reworked detrital material certainly have bypassed the basin and were deposited elsewhere. As discussed later, the sediment fill of the basin occurred during the rather short time interval of ca. 600 ka.

From particular primary volcanic intercalations in the series, it appears that some coeval volcanic centres (probably dykes) were situated near to the eastern margin. Their products such as banded rhyolitic lava flows (obsidian layers) are thought to travel only over distances of a few kilometres because of their high viscosity (Cas and Wright, 1987).

The basin-fill history in time and space can be summarised as follows (Fig. 5). More or less continuous sedimentation started with an eruptive phase giving rise to primary volcaniclastics evenly covering the basin floor (Infiernillo Mb., Fig. 5a). These mainly airborne volcanic deposits were reworked easily by small rivers and other gravity-induced surface movements. In the southern part and the northeastern margin, alluvial fan systems prograded into the basin (Ñamarin Mb.). The overlying fluvial system entered the basin from the northeastern margin. This bed- to mixed-load fluvial system (El Salado Mb., Fig. 5b) developed mainly along the NW border of the basin leaving coarse channelized deposits in the proximal part and finer braided river and swamp deposits in the distal, lower parts. The alluvial fans coming from the eastern margin appear to have been active during that time. This holds particularly for the south, because of interfingering with coeval swamp/anoxic lake (lignite-bearing) deposits of the El Salado Mb.

A short-lived syneruptive basin-fill episode is indicated by primary volcaniclastics (pyroclastic flows and falls) forming the base of the Dumapara Mb. These are overlain by sediments deposited from bedload-dominated braided river systems (Fig. 5c). The rivers had two main entry areas and converged in the southwest, occupying together the main surface of the basin. Notably, in the southernmost part the combined river system cuts through the distal parts of the older alluvial fan (Ñamarin Mb.) which formed a natural barrier at the basin outlet.

During the sedimentation of the Letrero Fm. tectonic activity seems to have decreased. In the central part of the basin a shallow lake was established (Fig. 5d). However, strong detrital input from small deltas (situated to the southeast, west and probably to the north) prevailed as documented by numerous coarse-grained intercalations into fine-grained lake sediments. Primary volcaniclastics are rare, diatomite layers are present in the very centre of the lake. Due to the detrital input the lake appears to have been filled very rapidly as evidenced in particular by the mixed-load meandering fluvial systems of the La Cruz Fm. entering from the central eastern and northern margin (Fig. 5e). This is inferred from the observation that this fluvial series has the greatest thickness where the central, deepest part of the former lake was situated.

The deposition of the Picota Fm. coincides with a eruptive activity occurring to the north or northeast of the basin, because the thickness of the volcaniclastic mass-flow wedge decreases rapidly towards the south (Fig. 5f). However, later erosion and removal in the southern part of the basin has accentuated this trend. After deposition of the Rumiloma Mb., the basin-fill series was partly eroded and incised, and the thin volcanic ashes of the Tambo Viejo Fm. sealed the resulting topography.

From the present analysis of facies and sediment distribution it becomes evident that only little input came from the western margin of the basin. This appears somewhat as a paradox because, as will be shown below, this margin was tectonically most active during the time of sedimentation. Obviously, the responsible NE–SW-trending marginal reverse fault zone formed an effective barrier sheltering the basin by deviating sediment transport towards the southwest.

5. Dating

5.1. Palaeomagnetism

Experimental approach

The palaeomagnetic samples were taken along three E–W profiles which stratigraphically cover

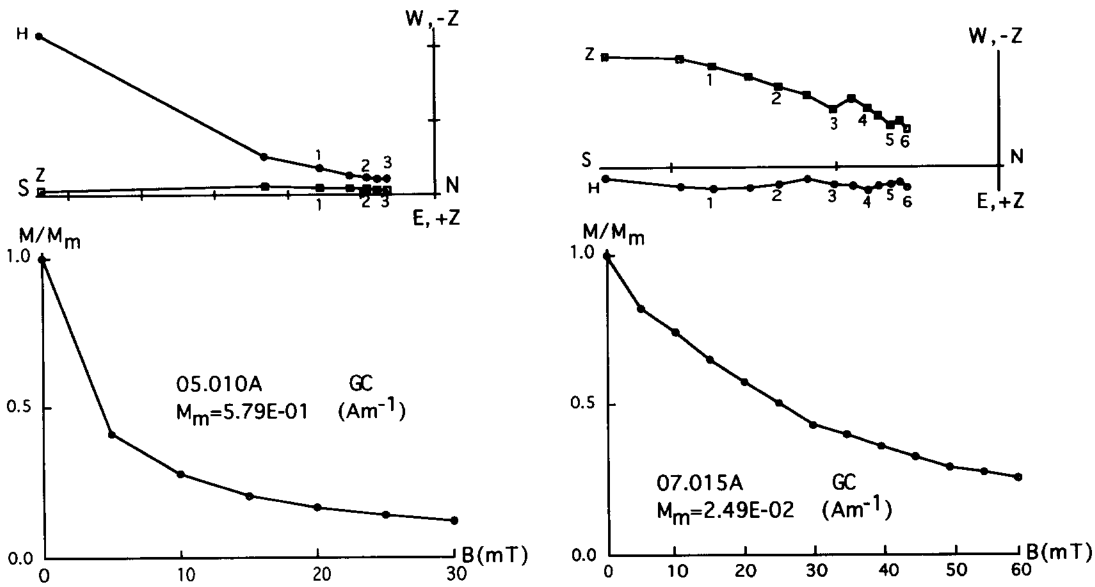


Fig. 6. NMR vectors (plotted on orthogonal vector diagrams in a geographical coordinate system) and intensities of the volcaniclastic sediments during AF treatment. Characteristic remanence (ChRM) components were determined by visual inspection and subsequent linear regression analysis of vector path segments trending towards or near the projection origin at elevated field strength.

the complete sequence from the Saraguro Fm. to the Patadel Mb. (Figs. 3 and 4). In addition, several sites were sampled outside the sedimentary basin in the basement volcanics of the Saraguro Fm. along the road from Cuenca to Loja. At each site three minicores, oriented with

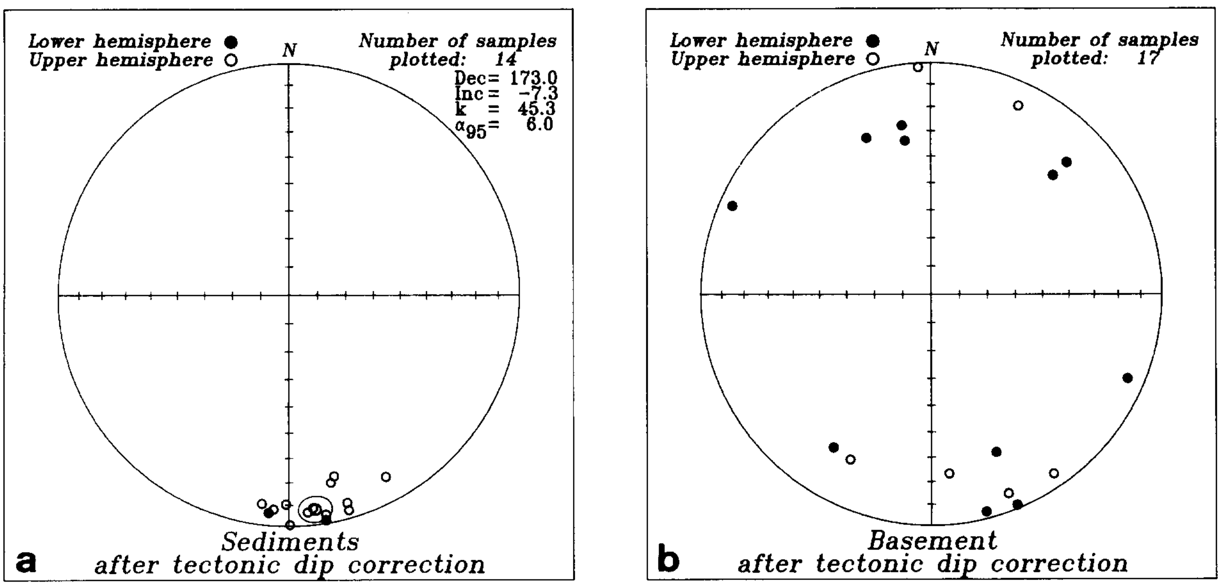


Fig. 7. AF cleaned characteristic remanence directions after tilt correction. (a) Mean direction of sediment-fill series including 95% circle of confidence; k = Fisher's (1953) statistical parameter. (b) Mean direction of basement sites (Saraguro Fm.). Number of samples refers to drill sites, each one comprising 3–4 measured samples.

a magnetic compass and inclinometer were drilled with a portable, gasoline-powered, drilling machine. The cores were cut into three to four one-inch specimens each for the palaeomagnetic measurements.

The samples were measured with a 3-axes 2G cryogenic magnetometer (Goree and Fuller, 1976). The instrument has a noise level of 5×10^{-6} A m^{-1} for one-inch samples if multiple readings (usually ten) are integrated in each of three different sample positions per measurement. All samples were demagnetised stepwise by alternating fields (AF) in 5 mT steps up to 30 mT or 60 mT, respectively. AF treatment often resulted in a substantial intensity loss of the natural remanent magnetization (NRM) after the first or second demagnetization step (Fig. 6: sample 05.010A). At higher fields, the NRM directions usually stabilized. The characteristic directions (ChRM) for each specimen were evaluated using linear regression techniques aided by visual inspection of the orthogonally projected NRM demagnetization paths. The ChRM vectorial mean directions for each site, including statistical pa-

Table 1
Palaeomagnetic data with mean declination and inclination of ChRM given together with Fisher's (1953) statistical parameters; grid references based on maps IGM (1970, 1971)

Site No.	Cores	NRM intensity	α_{95} (°)	k	Dec. (°)	Inc. (°)	Grid references
Basement							
1	3	3.45×10^{-1}	8.3	222.5	45.9	19.1	8.550/26.600
2	3	1.25	4.4	803.4	337.6	27.7	10.080/27.450
4	3	8.62×10^{-2}	8.0	236.9	145.7	−6.6	10.070/25.550
14	3	6.94×10^{-2}	6.6	353.9	157.8	1.8	12.700/24.880
17	3	1.48×10^{-1}	21.1	35.3	165.8	3.4	15.980/28.790
18	3	1.60×10^{-2}	6.5	364.9	294.0	6.4	13.460/31.060
19	3	8.18	15.0	68.8	113.3	8.2	12.610/31.600
28	3	9.10×10^{-1}	3.8	1081.1	212.6	22.3	16.300/35.180
31	3	7.30×10^{-1}	21.2	35.1	157.7	26.9	18.560/34.800
34	3	1.75	18.9	43.4	350.2	26.8	19.720/35.400
35	3	4.16×10^{-1}	18.1	47.4	350.4	33.1	21.950/32.940
36	3	5.66×10^{-1}	10.1	151.2	25.0	−11.1	21.380/33.950
37	3	9.59×10^{-2}	3.6	1181.2	206.2	−21.3	11.020/31.660
38	3	3.23×10^{-2}	15.1	67.7	356.8	−2.0	11.310/32.360
39	3	1.05×10^{-2}	6.4	375.4	45.8	27.0	10.580/32.500
40	3	1.28	15.5	63.9	174.3	−23.0	13.650/38.240
41	3	8.53×10^{-1}	4.1	922.8	158.8	−8.3	14.480/39.700
Sediments							
5	3	6.22×10^{-1}	13.1	89.9	187.4	−9.9	10.790/25.640
6	3	8.23×10^{-2}	6.9	320.9	184.1	−7.8	12.490/25.300
7	3	2.99×10^{-2}	6.5	356.8	170.4	−4.2	12.420/25.510
8	3	1.53×10^{-2}	16.4	57.5	164.4	−3.8	11.480/25.410
9	3	4.11×10^{-2}	20.4	37.8	164.3	−7.4	11.550/25.450
10	3	1.15×10^{-2}	17.2	52.7	151.9	−11.8	11.640/25.520
12	3	3.99×10^{-2}	3.8	1044.9	179.8	−0.7	11.250/24.950
15	3	2.13×10^{-1}	6.2	398.6	173.6	−8.0	12.680/24.950
16	3	6.44×10^{-2}	71.6	16.2	170.6	1.5	12.750/25.070
20	3	6.21×10^{-2}	7.1	298.4	167.4	−17.9	13.780/31.940
21	4	1.97×10^{-1}	7.9	137.3	166.0	−20.3	14.450/32.200
22	3	4.69×10^{-2}	8.2	227.1	180.9	−10.3	14.050/29.500
23	3	3.33×10^{-2}	10.6	136.1	175.1	−6.3	13.890/29.520
27	3	1.92×10^{-2}	8.9	192.1	185.5	6.0	16.860/28.150
Mean (n = 14)		1.05×10^{-1}	6.0	45.3	173.0	−7.3	

rameters (Fisher, 1953), are tabulated in Table 1 and plotted in Fig. 7.

Results

The mean NRM intensity of the sediments amounts to 0.105 A m^{-1} which is about one order of magnitude lower than the intensity of the basement volcanics. This high amplitude suggests a strong volcanogenic input of ferromagnetic minerals into the sediments. Curie temperatures around 570°C and the reversible behaviour of spontaneous magnetization during heating and cooling, measured for two samples from the Saraguro Fm. and the Dumapara Mb., point to the presence of rather pure magnetite in the sediments.

The ChRM directions of the sedimentary sites are very consistent (number of sites $n = 14$; $\alpha_{95} = 6.0°$) with a southerly mean declination $D = 173.0°$ ($\Delta D_{95} = \pm 6.04°$) and a shallow negative mean inclination $I = -7.3°$ (Fig. 7, Table 1). The basement sites, on the other hand, have low positive and negative inclinations, but largely scattered declinations so that a mean direction has not been calculated.

The basement samples are characterised by stable directional behaviour of NRM during AF cleaning and low within-site dispersion of ChRM directions (α_{95} of basement sites, Table 1). Apparently, both normal and reversed polarity states have been recorded in the basement sites which are distributed across more than 600 m of stratigraphic thickness in the Saraguro Fm. The variable declinations have to be explained by tectonic

Table 2
Fission-track analysis on zircons from pyroclastic horizons in the Saraguro Fm. (DH 31, MS 34, DH 68, MS 67, MS 24), the Nabón Group (MS 105, DH 98, WS 107, DH 94, DH 92, WS 68) and the Tambo Viejo Fm. (MS 100)

Sample	Grid ref.	Irrad. No.	No. of grains counted	ρ_d $\times 10^4$ cm^{-2} (counted)	ρ_s $\times 10^4$ cm^{-2} (counted)	ρ_i $\times 10^4$ cm^{-2} (counted)	$P\chi^2$ (%)	Bulk sample age $\pm 2\sigma$ (Ma)	Depositional age (*) $\pm 2\sigma$ (Ma)
DH 31	12250/29500	ETH-13-6	6	15.65 (1360)	295 (534)	275 (499)	26	28.2 ± 4.1	26.4 ± 4.5
MS 34	09820/25430	ETH-13-10	20	15.43 (1360)	295 (1322)	300 (1347)	< 2	26.7 ± 4.0	19.0 ± 3.5
DH 68	13480/31070	ETH-13-2	12	15.89 (1360)	331 (1054)	386 (1231)	41	22.9 ± 2.6	22.9 ± 2.6
MS 67	10450/27840	ETH-13-12	8	15.33 (1360)	293 (529)	337 (607)	9	22.5 ± 3.2	20.4 ± 3.1
MS 24	10365/24540	ETH-13-17	11	15.06 (1360)	174 (986)	201 (1139)	11	22.0 ± 2.6	18.5 ± 6.0
MS 105	11110/27870	ETH-22-17	9	13.93 (1090)	156 (522)	343 (1148)	8	10.7 ± 1.4	8.9 ± 1.3
DH 98	16300/28850	ETH-22-5	13	15.14 (1090)	124 (589)	319 (1512)	< 2	11.0 ± 2.0	8.3 ± 1.1
WS 107	16900/34775	ETH-22-3	11	15.34 (1090)	609 (337)	176 (974)	58	9.0 ± 1.4	9.0 ± 1.4
DH 94	16220/30170	ETH-22-6	11	15.04 (1090)	534 (357)	144 (963)	< 2	8.2 ± 1.3	7.5 ± 1.2
DH 92	16250/30880	ETH-22-8	12	14.83 (1090)	717 (261)	212 (771)	39	8.5 ± 1.4	8.5 ± 1.4
WS 68	17650/31725	ETH-13-14	10	15.22 (1360)	685 (201)	223 (653)	70	7.9 ± 1.4	7.9 ± 1.4
MS 100	13550/28200	ETH-22-2	10	15.44 (1090)	789 (283)	325 (1164)	73	6.3 ± 1.0	6.3 ± 1.0

ρ_d, ρ_s and ρ_i represent the densities of standard, spontaneous and induced tracks. $P\chi^2$ is the probability of obtaining χ^2 values for ν degrees of freedom where ν = number of crystals-1; mean ρ_s/ρ_i ratio used to calculate ages and uncertainty when $P\chi^2 < 5\%$. (*) marks ages using the method of Galbraith and Green (1990), statistically removing the detrital components. Grid references based on maps IGM (1970, 1971).

activity causing block rotation. The boundaries of these blocks, however, and the amount of rotation cannot be determined at present. Since the site inclinations are shallow throughout, main rotations by subvertical axes may be postulated. This would require fault zones with distinct horizontal displacement. Wether this type of tectonic activity caused the opening of the basin cannot be demonstrated with the presently available data. On the other hand, the rotations must have occurred prior to the sedimentary infill of the basin since the declinations of the sediments which are strictly antiparallel to the present geomagnetic field direction, do not give evidence of rotation after deposition.

All sediment sites show reversed ChRM polarity. Since the subhorizontal bedding stratification does not allow a fold test, the consistently reversed polarity is taken as evidence for an early NRM origin which has been imprinted during one chron of reversed polarity of the geomagnetic field. The fission-track data of the sediments result in a narrow limited age of 8 Ma. According to the geomagnetic timescale developed by Cande and Kent (1992), this age is characterised by the reversed chron 4r which is divided by the 30 ka normal subchron 4r.1n into two reversed subchrons 4r.1r and 4r.2r. The short normal subchron is not documented in our samples, either because the three profiles were not closely enough sampled or because sediments of that age do not outcrop; this short normal polarity interval will not be considered for the interpretation.

Sedimentation must have taken place during the short time span of chron 4r within about 640 ka at the longest. The resulting minimum sedimentation rate of nearly 1 m per 1000 yr is not extraordinarily high for volcanic areas (Vessel and Davies, 1981). The uppermost 250 m of sediments consist nearly exclusively of volcaniclastics (Picota Fm.) which could have been deposited in a very short time interval. No palaeomagnetic data are available for the youngest volcanic sediments (Rumiloma Mb.) because they are often weakly solidified or difficult to access. Their fission-track age of 7.9 ± 1.4 Ma, however, also falls within chron 4r.

5.2. Fission-track

Methods

Zircons were extracted from pyroclastic horizons (both air falls and flows) for fission-track dating; the sampling spanned the entire stratigraphic sequence and also the geographical extent of the basement. On examination of the heavy mineral suites, those with excessive detrital mineral components such as metamorphic mineral assemblages from the basement sequences were excluded.

The zircons were extracted using standard separation technique methods. The crystals were mounted in Teflon and polished. Etching was carried out at 210°C in a eutectic melt of KOH and NaOH (Gleadow et al., 1976) during periods varying from 36 to 72 h until tracks parallel and subparallel to the *c*-axis were well developed. Samples were irradiated together with a known age standard (Fish Canyon Tuff) and NBS glass dosimeters SRM 612. All ages were determined using the zeta approach (Hurford and Green, 1983). This value was 338 ± 5. The errors are those recommended by Green (1981) and are reported at the 2σ level (Table 2).

Results

The twelve ages are the first fission-track analyses to be reported from the continental sediments of the Sierra of Ecuador (Table 2, Fig. 4). Together with the palaeomagnetic results they present a uniform picture. Some of the zircons in these horizons are clearly detrital and the younger component only represents the age of volcanism. This age (depositional age, Table 2) was determined using the technique recommended by Galbraith and Green (1990).

(1) The Saraguro Fm. ranges in age from 26 to 19 Ma (Fig. 4). These ages are younger than those reported by Lavenu et al. (1992, 35.3–26.8 Ma). Inspection of the radial plots in three of the samples hints at a detrital component which was removed statistically. The ages of 22.9 ± 2.6, 20.4 ± 3.1 and 19.0 ± 3.5 Ma support the idea of Kennerley (1980, 21.4 Ma) that the Saraguro Fm. is Early Miocene.

(2) Most of the basin fill (Nabón Group) was formed in a very short period between 8.5 and 7.9 Ma (Late Miocene, Chron 4r). The Nabón Group may be considered the time equivalent to the top of the Mangán Fm. in the Cuenca basin (Fig. 2 and Lavenu et al., 1992). The discordance between the basement and the Nabón Group represents a hiatus of about 11 Ma. At the southwestern edge of the basin only, some older volcaniclastics lie unconformably on the basement; they were dated at 18.5 ± 6.0 Ma (Early Miocene). These probably represent volcaniclastics which were intercalated within the Saraguro Fm. (Baldock, 1982) and exhumed by later erosion. Their age is in good agreement with the youngest ages of the Saraguro Fm.

(3) The youngest volcanic activity in the region of Nabón, as recognized by the air fall deposits of the Tambo Viejo Fm. (Pisayambo Fm., Litherland et al., 1993) lie discordantly on the basin fill and on the Saraguro Fm. They are dated at 6.3 ± 1.0 Ma (Late Miocene). These sediments document the youngest pyroclastic deposits of the southern Sierra of Ecuador south of 2°S (Barberi et al., 1988; Lavenu et al., 1992).

$^{40}Ar-^{39}Ar$ dates (Madden, 1990) of the middle and upper basin fill are 10.85 and 10.5 Ma, respectively. Such ages fall in the zone of normal polarity, chron 5n (Cande and Kent, 1992). However, the whole basin filling has reversed polarity and therefore cannot have been deposited during chron 5n. The difference between these ages and the present ones may be explained by the fact that these $^{40}Ar-^{39}Ar$ determinations are done on bulk mineral samples from grouped horizons, whereas the fission-track method is a single-crystal technique on minerals from single beds. Detrital contamination is observed in the fission-track through examination of the radial plots, especially DH 98 (Fig. 8 and Table 2) which yield initial ages similar to those of $^{40}Ar-^{39}Ar$. However, by splitting the samples statistically (Galbraith, 1990) the older detrital ages are removed. Therefore, on this basis and on the evidence of the palaeomagnetics we believe that the fission-track data are more accurate. The younger component represents the volcanic event.

6. Structures and synsedimentary tectonics

The Nabón basin is characterised by mainly flat-bedded units, locally affected by syndepositional contractional structures such as growth folds, growth faults and cumulative wedges. Such structures have been well described from foreland basin environments (Riba, 1976; Anadón et al., 1986) and from other intermontane basins of the Andes such as the Cuenca basin (Noblet et al., 1988).

Sedimentary units deposited during vertical tectonic movements develop wedge-shaped geometries with visible basinwards fanning (thickening) of the layers (Miall, 1978). In the Ebro basin of NE Spain such wedges were also called progressive unconformities (Riba, 1976; Anadón et

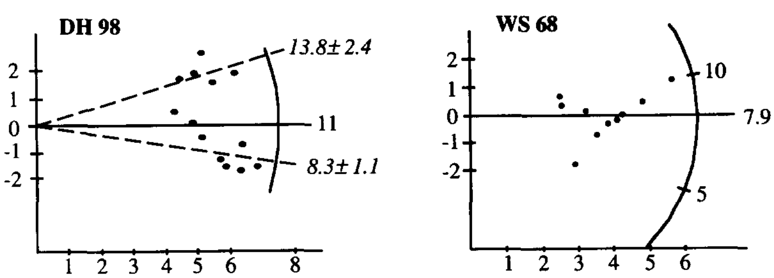
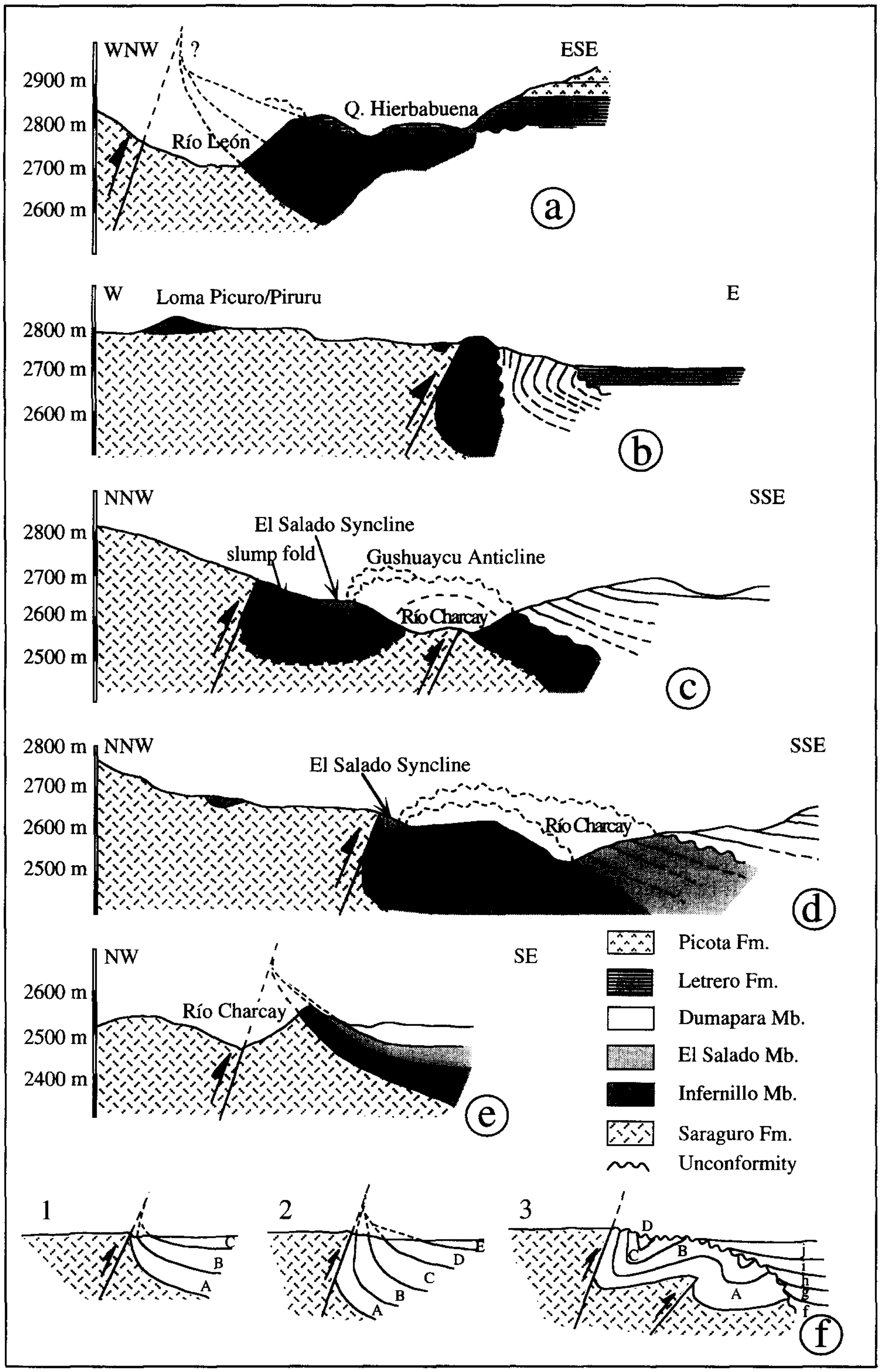


Fig. 8. Radial plots (after Galbraith, 1990) of the samples DH 98 and WS 68 (Nabón Group) showing multiple- and single-sample crystal populations. On the x-axis (drawn horizontal) the reciprocal error is plotted on a linear scale, while the y-axis (drawn vertical) shows the standardised age estimate. The curve displays ages in Ma. The single-crystal ages of DH 98 (Table 1) can be divided into two populations, shown with dashed lines (after Galbraith and Green, 1990). The single-crystal ages of WS 68 (Table 1) are all from one population falling within the field of $y = \pm 2$, yielding a mean age of 7.9 ± 1.4.



WNW
?
ESE
2900 m
2800 m
2700 m
2600 m
Q. Hierbabuena
Río León
a
W
Loma Picuro/Piruru
E
2800 m
2700 m
2600 m
b
NNW
SSE
El Salado Syncline
slump fold
Gushuaycu Anticline
Río Charcay
2800 m
2700 m
2600 m
2500 m
c
2800 m
NNW
SSE
El Salado Syncline
Río Charcay
2700 m
2600 m
2500 m
d
NW
SE
2600 m
2500 m
2400 m
Río Charcay
e
Picota Fm.
Letrero Fm.
Dumapara Mb.
El Salado Mb.
Infernillo Mb.
Saraguro Fm.
Unconformity
1
2
3
A
B
C
D
E
f

al., 1986). However, this term is less appropriate and will not be used here, because in proper use it should refer generally to the geometry of sediment bodies which might or might not bear internal unconformities. Therefore, for composite systems of wedges bearing internal unconformities the term cumulative wedge (Riba, 1976) is more appropriate. The lateral and vertical shift of the depositional axis can be depicted by on- and offlap relations of the wedges with respect to the position of the generating tectonic structure, and coeval deformation of wedges can be described by the orientation of the rotation axis of the wedge layers.

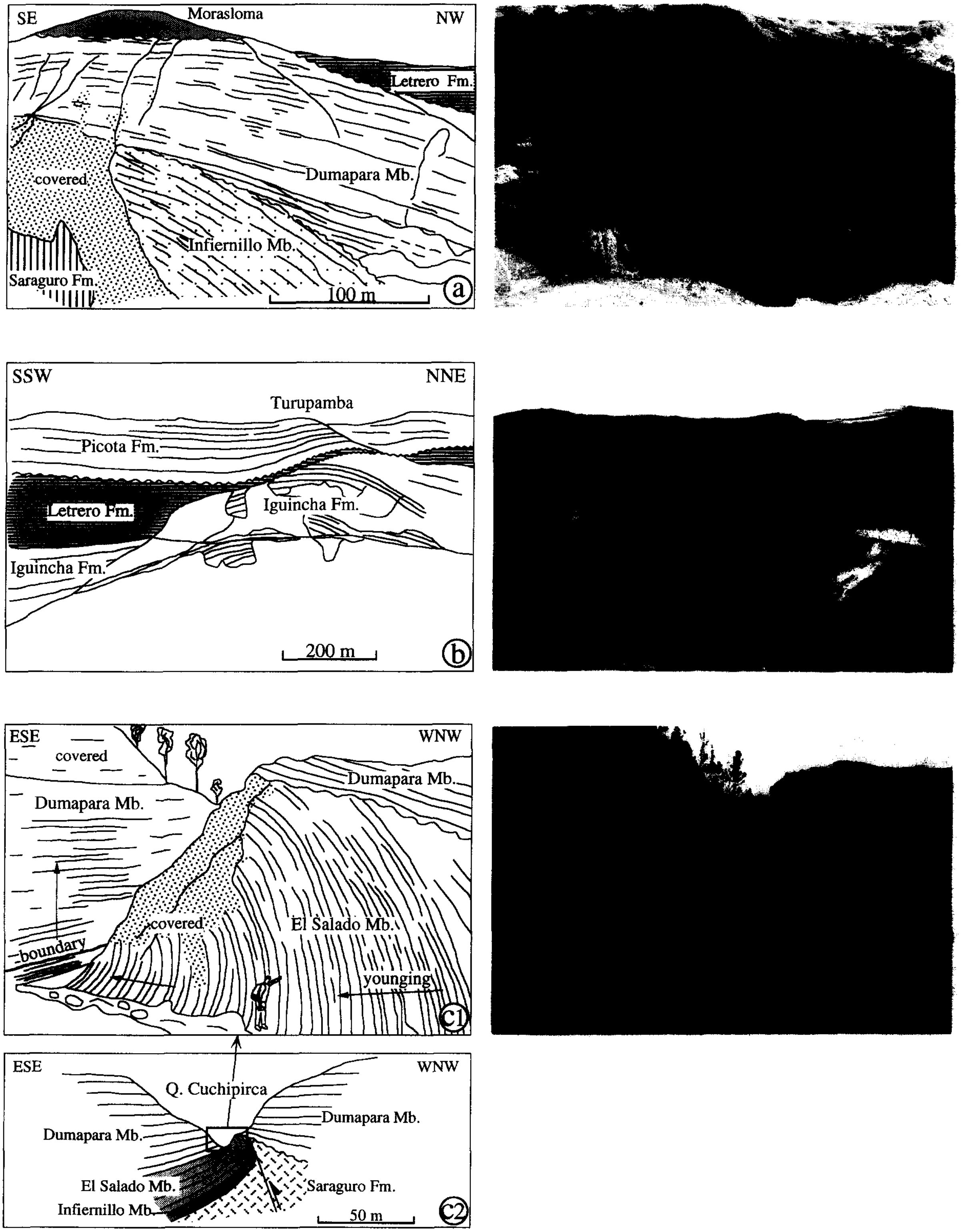
In the Nabón basin the deformation history can be subdivided into four phases based on the age of the affected sediment-fill formations (Fig. 3). These stages are: pre-fill, early and late fill phases and a post-fill deformation phase. The early and late fill stages are separated by a period of minor tectonic activity during the deposition of the Letrero and La Cruz Fms. However, it is observed that the same structures were reactivated after deposition of the Letrero Fm. Therefore, the present distinction between early and late fill phase deformations is guided by the age of the youngest sediment series apparently involved in deformation. The early fill phase deformation is most pronounced and can be seen best in the western half of the basin. During this stage mainly the Iguincha Fm. and to a minor extent the base of the Letrero Fm. were deformed. Late fill phase deformation can be observed in the eastern central and northern part of the basin and is manifested mainly by long-wave deformation features in the Picota Fm. Pre-fill phase tectonic movements are documented in the southern part of the basin at Loma Garusario and near the cemetery of Cochapata where a moderate NW–SE-striking horst–graben system developed in the Saraguro Fm. basement series which was filled and overstepped by deposits of the Infiernillo and Ñamarin Mbs.

Within the Nabón basin, cumulative wedges were often formed and deformed by differential uplift along basement block faults. These reverse faults probably developed along pre-existing anisotropies in the Saraguro Fm. caused by subvertical cooling columns. During narrowing of the sedimentation space, growth folds (syn- and anticlines) also developed (Fig. 3). Due to the deeply incised river valleys numerous dramatic examples of structures are exposed on cliff faces which can be correlated over distances of several kilometres. In the following we will discuss some of these features.

6.1. The western basin margin

Important early fill stage deformation is recorded along the western margin of the basin where a major NW-dipping reverse fault has uplifted the basement. Along much of this margin there is a clear faulted contact between basement and the basin infill. The intensity of deformation changes along-strike as shown in a series of cross-sections (Fig. 9 and locations in Fig. 3). In the northernmost part of the basin the fault is sealed by the Picota Fm. but a wedge in the underlying Infiernillo Mb. northwest of Shiña indicates its existence. In central and southern regions a generally N040°-striking reverse fault uplifting the basement to the west with respect to the sediment-fill series can be well constrained. In a central position in the area of Sigüir, the sediments of the Infiernillo Mb. are steeply dipping and overturned along the western margin, showing additional small-scale faults, internal un-

Fig. 9. Profiles through the western margin of the Nabón basin arranged from N (a) to S (e). Location of profiles shown in Fig. 3. The western margin is characterized by a major early fill phase reverse basement fault causing significant deformation and wedging (cumulative wedges) in the Iguincha Fm. See text for discussion. (f) This shows an idealised succession of reverse fault activity and sedimentary wedge development. Stage *1* displays offlap structures: profiles (a) and (e); stage *2* documents wedge growing, indicating ongoing tectonic activity: profile (d); stage *3* shows the development of a second reverse fault which generates folding and unconformable onlap of sediments (capital letters) over the older structure: profile (c). Major unconformities between lithostratigraphic formations and members are indicated by wavy lines.


SE
Morasloma
NW
Letrero Fm.
Dumapara Mb.
covered
Infiernillo Mb.
Saraguro Fm.
100 m
a
SSW
NNE
Turupamba
Picota Fm.
Letrero Fm.
Iguincha Fm.
Iguincha Fm.
200 m
b
ESE
WNW
covered
Dumapara Mb.
Dumapara Mb.
covered
El Salado Mb.
boundary
younging
c1
ESE
WNW
Q. Cuchipirca
Dumapara Mb.
Dumapara Mb.
El Salado Mb.
Saraguro Fm.
Infiernillo Mb.
50 m
c2

conformities and ordered growth folds (axes are oriented N040°, N020°, Figs. 9b, 9c). Here the boundary between basement and basin fill becomes diffuse and the marginal fault is probably split and bent westward (Fig. 3). Further to the south the layers are less steeply inclined (Fig. 9d) and thick sedimentary wedges in the Iguincha Fm. open towards the basin interior to the east (Figs. 9d, 9e). The Infiernillo and Dumapara Mbs. series commonly show offlap geometries indicating that uplift was faster than sediment fill. The Letrero Fm. sediments show onlap geometries (e.g. Fig. 9b and examples in Fig. 10) indicating a decreasing of uplift along the margin. Local and regional synsedimentary uplift caused considerable relief and sedimentary/erosional breaks (note missing El Salado Mb. in Figs. 9b, 9c). In the region of SW Loma Tambo Viejo (Fig. 3) a set of early fill stage normal faults trending perpendicular to the major marginal reverse fault is recognized. Sedimentary wedges have developed in the hanging-walls indicating a syn-Iguincha Fm. extensional activity.

We conclude that the synsedimentary reverse fault at the western basin margin was strongly active (ca. 200 m upthrow) during the early basin-fill history up to the time of deposition of the Letrero Fm. The fault zone is interpreted as dipping steeply to the northwest. The throw was variable along the reverse fault. This caused different deformation patterns (weak in the north and south, intense in the centre). However, we think that the fault geometry has been slightly modified by later movements.

6.2. The eastern basin margin

The contact between basement and basin sediment fill can only be observed in the southernmost part of the eastern basin margin. The presence of uplifted basement blocks near the margin (e.g. Sucurumi and Morasloma areas) supports the existence of a major tectonic line. The overlap of the contact by young volcanic ash deposits (Tambo Viejo Fm.) indicates that fault activity ceased, at the latest, after the sedimentation of the Picota Fm.

6.3. Basin interior structure

Morasloma

In the basin interior there are several other major structures which were produced either by differential basement uplift or by synsedimentary reverse faults. Two such features are situated in the central and northern part (Fig. 10). The early to late fill phase structure of the Morasloma area is located on the SW flank of a N030°-striking basement block (Fig. 3). During uplift a thick cumulative wedge (ca. 200 m thickness measurable in the field) developed, opening towards the northwest. Within this wedge three unconformities between the Infiernillo Mb. (the Dumapara Mb., the Letrero Fm. and the Picota Fm.) can be observed. Notably, at the Morasloma culmination the Letrero and La Cruz Fms. are lacking (Fig. 10a). To the east of the basement uplift, a parallel growth syncline was formed in the Iguincha Fm. The Morasloma basement uplift continues with a ca. 10° plunge towards the southwest and can be recognized there as an open anticline, deforming beds up to the base of the Picota Fm. (Fig. 3). To the northwest the Morasloma structure is paralleled by a similarly plunging gentle syncline extending from La Playa in the direction of Nabón village. Both these folds are thought to represent early to late fill phase deformation.

Fig. 10. Variably scaled examples of synsedimentary deformational features as depicted by the geometry of bedding in the basin-fill series. For location of photographs see Fig. 3. Major unconformities are indicated by wavy lines. (a) Morasloma early fill phase structure: uplift of the Saraguro basement and cover (Infiernillo Mb.) is followed by onlap and overstepping by the Dumapara fluvial sediments with prominent opening of the sedimentary wedge towards the northwest. The lake deposits of the Letrero Fm. fill a synclinal depression to the northwest. Evidently, the Morasloma culmination represented an important positive relief feature, because it is overlain paraconformably by the volcaniclastics of the Patadel Mb. (b) Turupamba late-phase fill structure: similar situation as before (a). The lake sediments of the Letrero Fm. fill the depression around the uplifted Iguincha Fm. However, continued uplift is responsible for the development of a SW opening wedge in the Picota Fm. (c) Quebrada Cuchipirca early fill phase structure: the general younging of beds occurs in ESE direction being folded around a NE–SW-trending fold axis. Younger beds of the Dumapara Mb. discordantly overlie the synsedimentary structure.

Turupamba

A comparable situation can be found in the Turupamba area in the eastern central part of the basin (Fig. 3). There the basin-fill series up to the Patadel Mb. (Picota Fm.) is involved in a large growth anticline striking N015° and plunging gently to the north-northeast. In the western limb over a distance of about 1 km a pronounced onlap of the Letrero Fm. on the Iguincha Fm. and fanning of the Picota Fm. is clearly visible (Fig. 10b). The Turupamba structure can be correlated along-strike with the basement uplift in the Sucurumi area where an important wedge in the Picota Fm. opens westwards (Fig. 3). Both structures in the Morasloma and Turupamba areas cannot be linked directly in space, but their similar development appears evident. Related synsedimentary deformation and fanning features present in the Picota Fm. qualifies the Turupamba structure as a late-stage fill feature.

Cuchipirca

A very dramatic small-scale example of an early-phase cumulative wedge is present in the Quebrada Cuchipirca in the SW part of the basin (Fig. 10c). The sedimentary wedge in the Infiernillo, El Salado and Dumapara Mbs. opens in an eastward direction and has rotated around a NE–SW-trending axis. From the base of the river upwards to the right the vertical layers have been bent into a 45° overturned position, facing more or less west-northwest. The rotation axis of the sediment wedge strikes N050° or parallel to the basin's western margin. A local unconformity between the El Salado and Dumapara Mbs. is well visible in the upper right part of the photograph. From the stratigraphic content of the wedge and the deformation style it appears that this structure probably formed coevally with the main thrust fault along the nearby western margin in Figs. 3 and 9.

From the orientation of the synsedimentary reverse faults, uplifted basement blocks, folds and sedimentary wedges, it is clear that sedimentation was accompanied by NW–SE-directed shortening. Few synsedimentary normal faults (Fig. 3, Loma Tambo Viejo area) document the extension in NE–SW direction. Post-fill tectonic structures such as normal faults and landslides can be detected on aerial stereo-photographs (IGM, 1976) and are generally oriented perpendicular to the basin axis and therefore point to continued NE–SW-directed extension after sedimentation ceased (Fig. 3). The preservation of the ruling shortening direction after sedimentation is also implied by the fact that the present drainage direction (slope orientation) is still similar to that during the deposition of the Nabón Group. This means that the orientation of the principal strain field in the area around Nabón did not change significantly after basin fill and has probably persisted up to the present day. However, the rapid onset of postsedimentary erosion points to accelerated regional uplift of the basin area.

7. Conclusions and discussion

The history of the intermontane Nabón basin of Ecuador can be summarised as follows:

(a) The sediment-fill series reflects variations in the volcanic activity in nearby volcanic centres: syneruptive fill stages consist of primary volcaniclastics, while intereruptive fill stages are characterised by aquatic reworking and deposition of volcanic sediments (fluvial, alluvial and lacustrine formations).

(b) Basin infill occurred during the short period from 8.5 to 7.9 Ma (Late Miocene, 4r chron) and the series is bound by major unconformities between the lower ignimbrite basement and overlying younger volcanic ash fall deposits. The earlier time gap is about 11 Ma, the latter 1.5 Ma.

(c) The orientation of synsedimentary faults, folds and cumulative wedges indicates that WNW–ESE maximal shortening occurred during the infill period, i.e. perpendicular to the longer basin axes.

The present data demonstrate that the history of the intermontane basin evolution in Ecuador is more complex than hitherto thought. In a regional stratigraphic correlation the Nabón Group represents a time equivalent of the top of the Mangán Fm. in the Cuenca basin. The temporal and sedimentary evolution of these two basins are therefore fundamentally different.

The southern Ecuadorian intermontane basins are interpreted to have formed as pull-aparts along major crustal faults related to a general right lateral tectonic regime between the Cordilleras Occidental and Real (Lavenu et al., 1990, 1993). Early Miocene basin opening is thought to have occurred along N–S strike-slip faults and normal faults oriented N020° to N040°, the latter parallel to the long axes of the Nabón basin. Middle to Late Miocene basin fill occurred during N060° and N080–N130° shortening. The reconstruction of the syndepositional strain fields in the different Early/Middle Miocene basin-fill series as given in Lavenu et al. (1990) cannot be applied to the Nabón basin, because there is simply no sediment record preserved from that time. The NW–SE-oriented maximal shortening revealed in the Nabón basin fill is comparable with the Late Miocene stress field proposed for the Cuenca basin by Noblet et al. (1988). However, in the Nabón basin-fill series we found no clear structural evidence for synsedimentary strike-slip movements. Our palaeomagnetic data would suggest block rotations in the underlying basement, probably due to transcurrent movements. If translational block movements did occur, after the present stratigraphic evidence they must have occurred before basin infill. Given that the Nabón basin would have been situated on or along a major strike-slip fault, the observed strain regime in the sediments is incompatible with that expected from classical strike-slip theory where the direction maximum of horizontal compression makes an angle of 30–45° with the main vertical fault (Wilcox et al., 1973). But it also appears possible that the Nabón basin and other intermontane basins were situated in the central axial parts of a regional translational system, near the weak fault, showing the deviation of the shortening direction perpendicular to the main displacement axes as proposed by Zoback et al. (1987).

The timing of the opening of the Nabón basin remains unclear. There is no evidence of volcanic activity between 19 and 8.5 Ma. This can be explained either by erosion during the early stage of basin-forming tectonics or that the basin only began to fill at about 8.5 Ma and there was no significant volcanic cover laid down before. Radiometric age determinations in the volcanic deposits of southern Ecuador (15.4, 12.2, 11.2 Ma, Barberi et al., 1988, and 14.2 Ma, Kennerley, 1980) suggest a more or less continuous Miocene volcanic activity. However, there is also no evidence that the volcanic products were equally distributed over the whole area. Thus, strong local uplift could have triggered important erosion of coeval volcanics. For the moment we are inclined to assume that the Nabón basin structure was formed with the onset of sediment infill.

The uplift history of the Ecuadorian Andes and the effects on the intermontane basin formation is poorly constrained stratigraphically. The Interandean depression could have formed during the Late Miocene (Baldock, 1985). In Peru and the Central Andes a general tectonic model for the Neogene consisting of several contractional pulses was proposed (e.g. Sébrier et al., 1988; Noble et al., 1990). In comparison, the Nabón basin fill and deformation coincides more or less with dated discordances, contractional deformations and palaeovalley-fill events in the Colombian and Peruvian Andes dated at about 8–10 Ma (Quechua-II compressive pulse of Sébrier et al., 1988 and Noble et al., 1990). This phase is assumed to have been preceded by strong regional uplift since the Quechua-I pulse calibrated as Early Miocene. This supposed uplift period correlates with an accelerated subduction rate of the Nazca plate from 20 to 10 Ma (Pardo-Casas and Molnar, 1987; Daly, 1989). Related strong regional uplift in the Nabón area could have been responsible for the observed basal discordance from about 19 to 8 Ma. Preliminary pollen and spores data from the Nabón basin series point to the presence of a montane rain forest environment at a palaeoaltitude of around 1000 to 1500 m at 8 Ma (Schatz, 1994). These numbers agree with an apparent continuous uplift of the Ecuadorian Andes since their emergence in the Late Eocene (Kennerley, 1980) to reach the present mean elevation of the Nabón area at 2500 m. The younger sedimentary hiatus (7.9–6.3 Ma, combined with an angular unconformity) between the Picota Fm. and the Tambo Viejo Fm. correlates probably with the regionally recognized contractional Quechua-III pulse

(Sébrier et al., 1988; Noble et al., 1990) dated at about 7 Ma.

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